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(12) **United States Design Patent**  
**Ellegaard**

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(54) **FREQUENCY CONVERTER**

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(\*\*) Term: **15 Years**

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Jan. 6, 2022 (EM) ..... 008821797

(51) **LOC (15) Cl.** ..... **13-02**

(52) **U.S. Cl.**  
USPC ..... **D13/110**

(58) **Field of Classification Search**

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D13/199; D15/5, 149, 199

CPC ..... H02M 7/00; H02M 7/02; H02M 7/003;  
H05K 5/00; H05K 5/0017; H05K 5/04;  
H05K 5/0217; H05K 7/20; H05K  
7/20136; H05K 7/20209; H05K 7/20418;  
H05K 7/1432; G06F 1/20; E04H 15/40;  
E04H 15/58; E04H 15/405

See application file for complete search history.

(56) **References Cited**

**U.S. PATENT DOCUMENTS**

D821,324 S \* 6/2018 Viman ..... D13/123  
D913,920 S \* 3/2021 Tachibana ..... D13/110  
D935,405 S \* 11/2021 Andersson ..... D13/110  
D939,441 S \* 12/2021 Tachibana ..... D13/110  
D988,996 S \* 6/2023 Huhtakallio ..... D13/110

D1,009,791 S \* 1/2024 Huhtakallio ..... D13/110  
D1,020,641 S \* 4/2024 Soerensen ..... D13/110  
D1,021,792 S \* 4/2024 Soerensen ..... D13/110  
D1,021,793 S \* 4/2024 Soerensen ..... D13/110  
D1,021,794 S \* 4/2024 Soerensen ..... D13/110  
D1,021,795 S \* 4/2024 Soerensen ..... D13/110  
D1,021,796 S \* 4/2024 Soerensen ..... D13/110  
D1,028,887 S \* 5/2024 Soerensen ..... D13/110  
D1,028,888 S \* 5/2024 Soerensen ..... D13/110  
2001/0019495 A1 \* 9/2001 Miettinen ..... H02M 7/003  
363/157  
2008/0084148 A1 \* 4/2008 Talja ..... H05K 7/1432  
312/223.6  
2012/0257338 A1 \* 10/2012 Silvennoinen ..... H02M 5/00  
361/679.01  
2012/0272494 A1 \* 11/2012 Markeson ..... F16M 7/00  
29/402.01

\* cited by examiner

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(57) **CLAIM**

The ornamental design for a frequency converter, as shown  
and described.

**DESCRIPTION**

FIG. 1 is a front, right side and top perspective view of a  
frequency converter comprising a new, original and orna-  
mental design;

FIG. 2 is a front view thereof;

FIG. 3 is a rear view thereof;

FIG. 4 is a top view thereof;

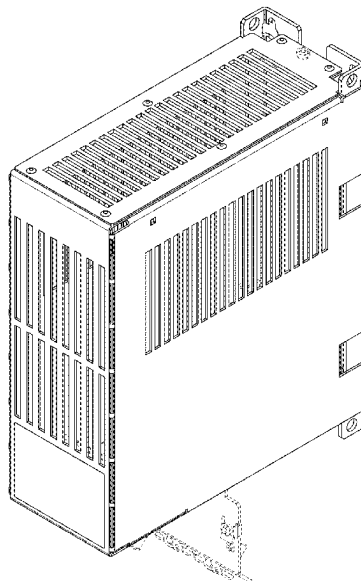
FIG. 5 is a bottom view thereof;

FIG. 6 is a left side view thereof; and,

FIG. 7 is a right side view thereof.

The broken lines illustrate portions of the frequency con-  
verter that form no part of the claimed design.

**1 Claim, 7 Drawing Sheets**



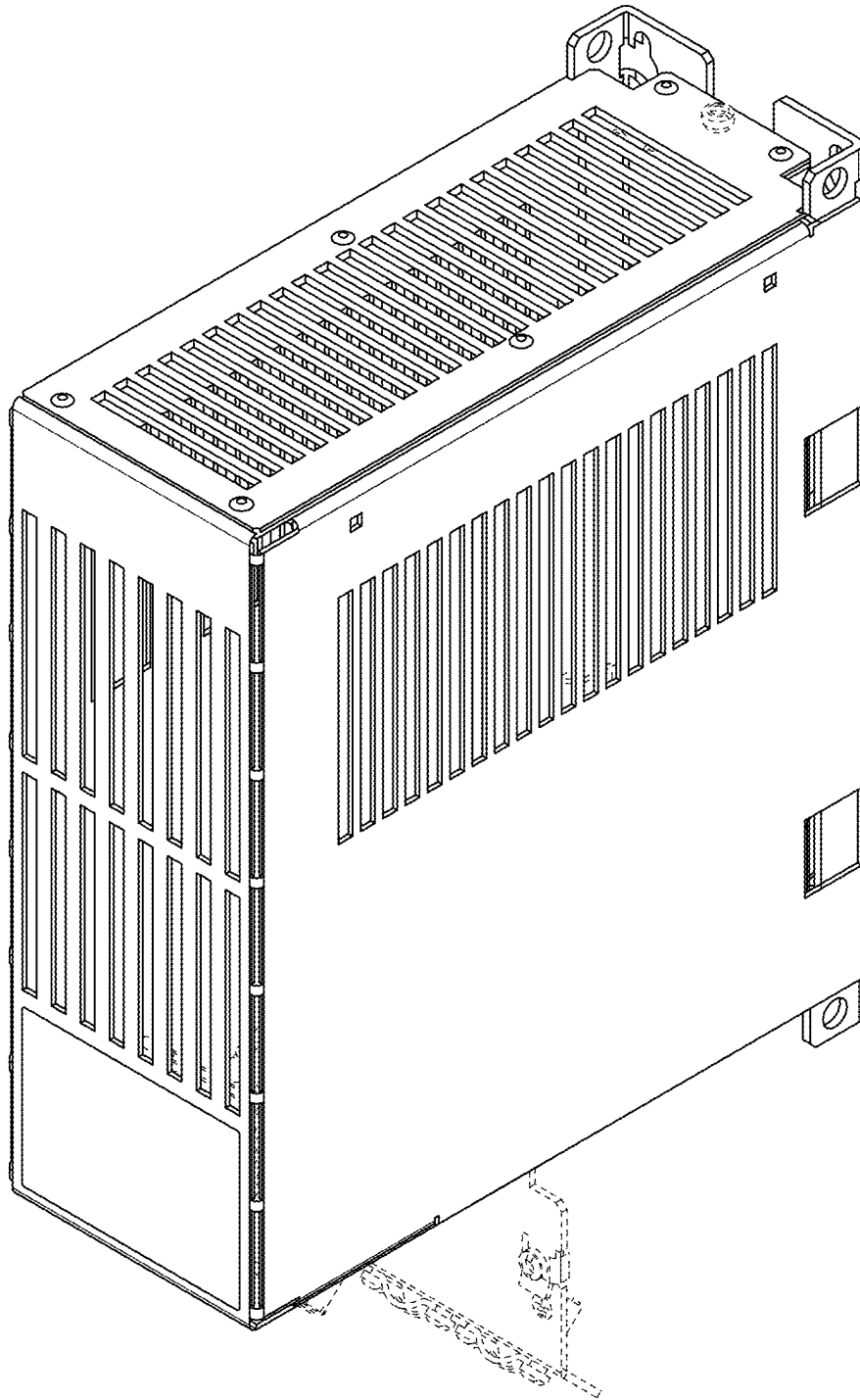


FIG. 1

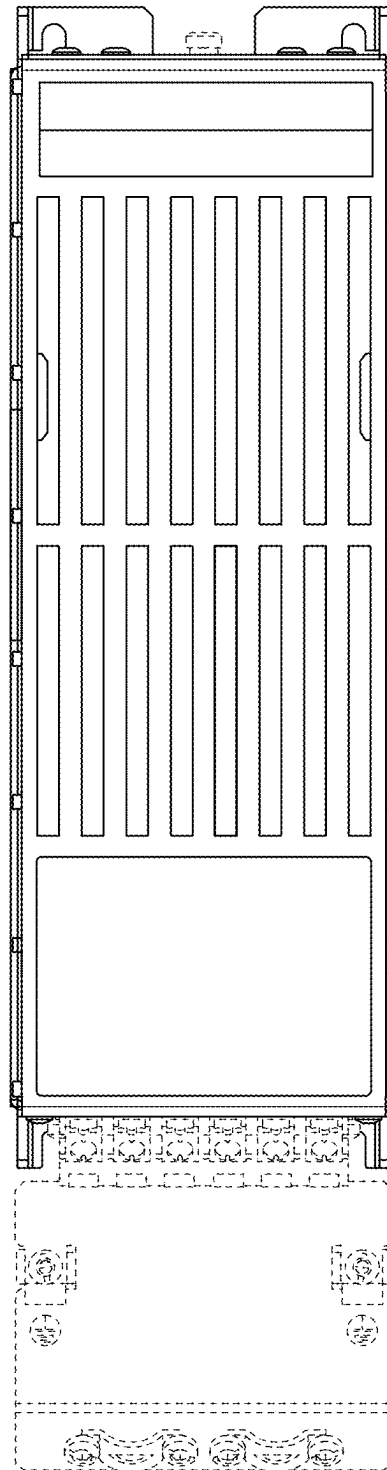


FIG. 2

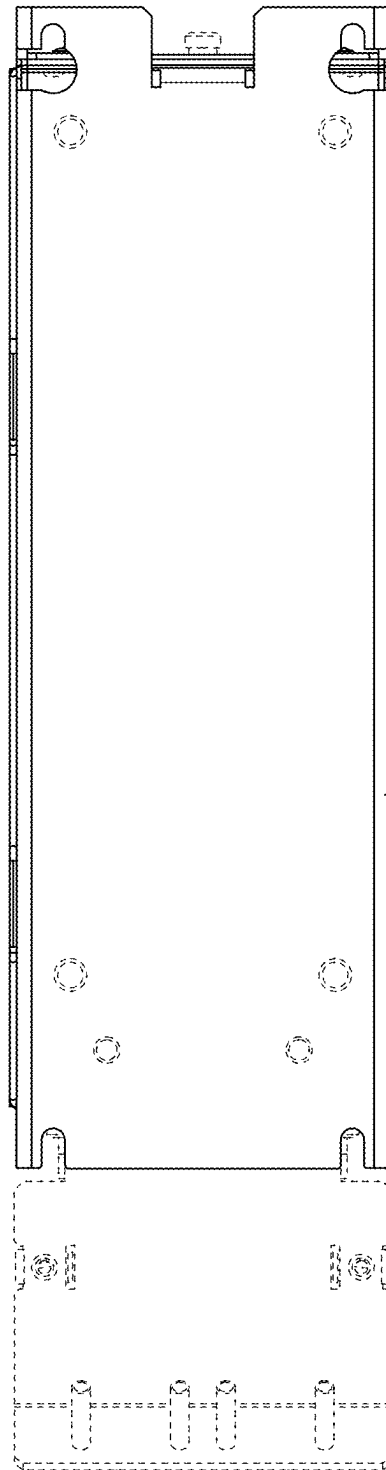


FIG. 3

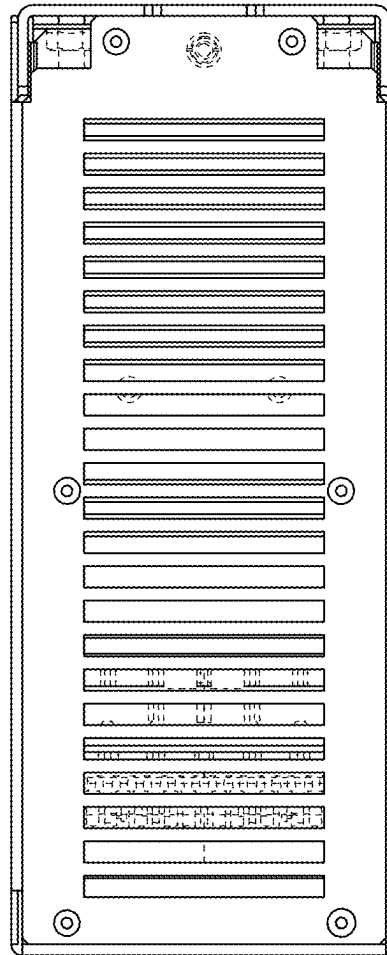


FIG. 4

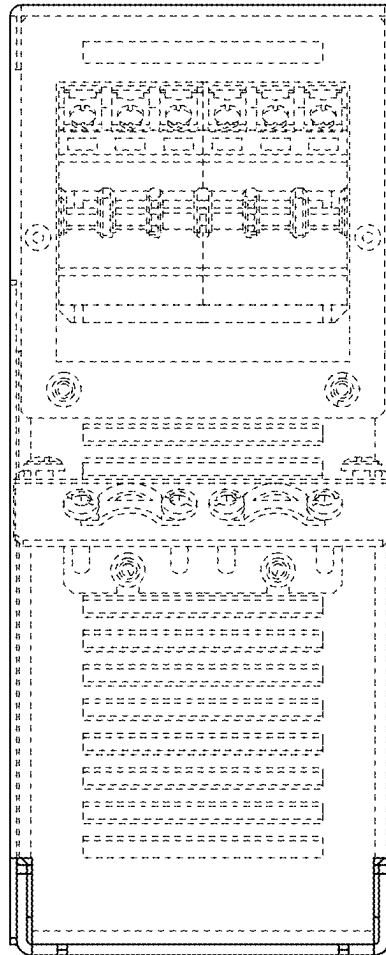


FIG. 5

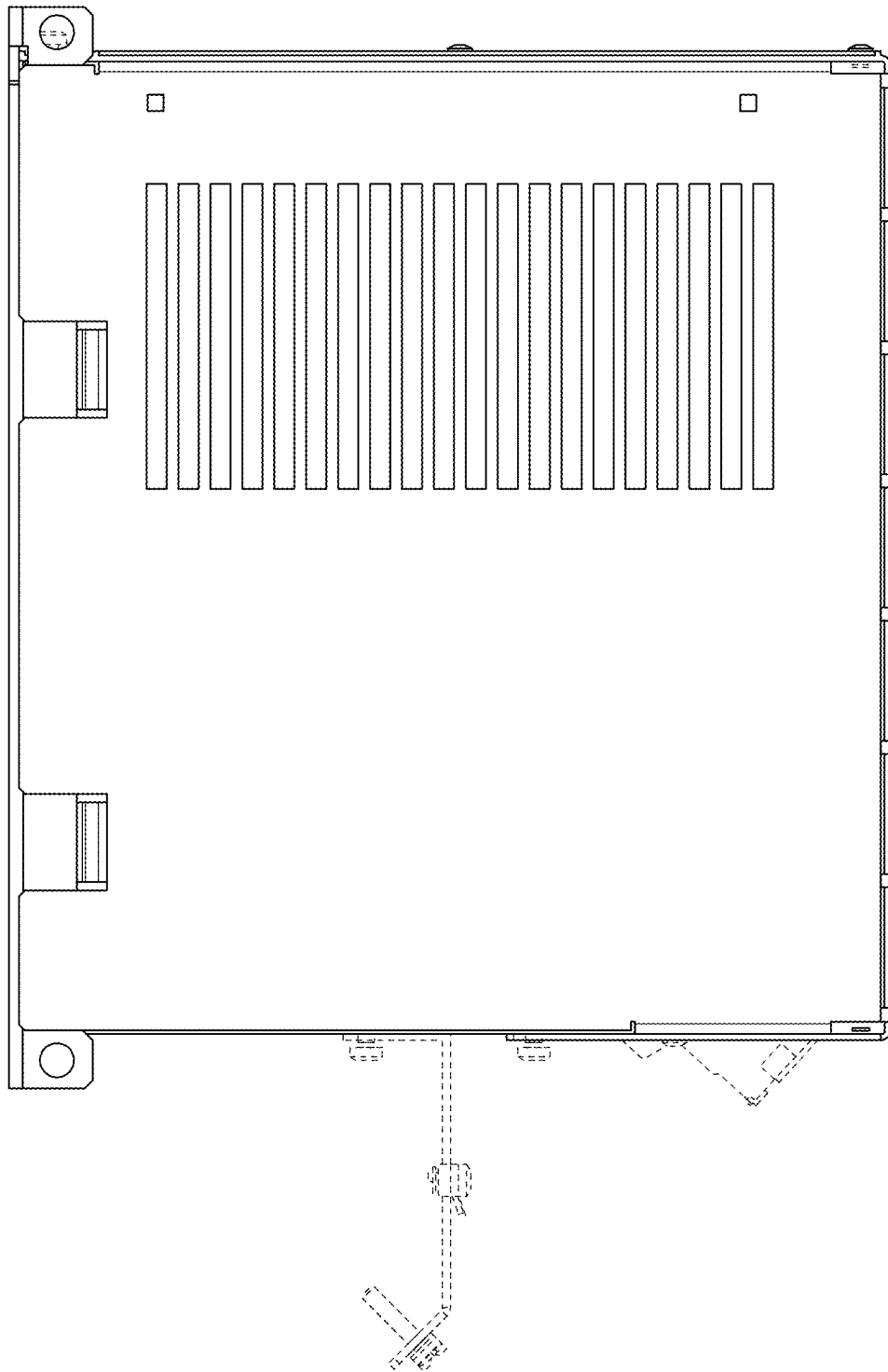


FIG. 6

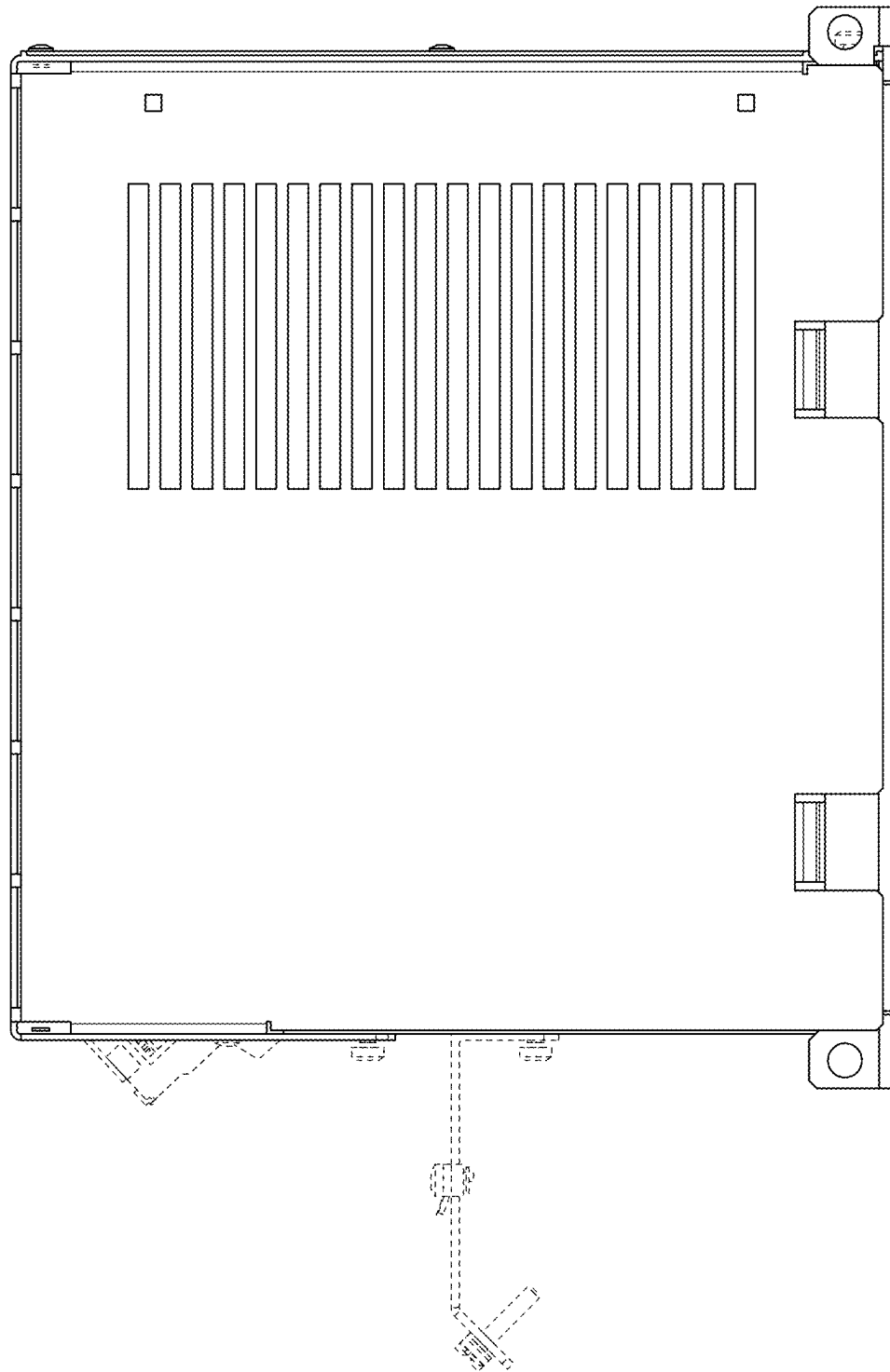


FIG. 7